

A MAJOR PROJECT SYNOPSIS

ON

FabMetrics AI

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING



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CERTIFICATE

This is to certify that the Major Project topic “**FabMetrics AI**” submitted by **Chitransh Saxena (9923102040)**, **Dhruv Kumar (9923102051)**, and **Shubhangi Mathur (9923102054)** is new, appropriate and not repeated/copied from previously submitted project works.

Signature of Supervisor:

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DECLARATION

We hereby declare that the title of the Major Project, “**FabMetrics AI**,” is not repeated/copied from previously submitted project works and that we have not misrepresented, fabricated, or falsified any idea, data, fact, result, or source in our submission.

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PROBLEM STATEMENT

Semiconductor wafer-map inspection is an important stage of manufacturing quality control because spatial defect patterns can indicate process, equipment, or handling abnormalities. Manual inspection of large numbers of wafer maps is time-consuming and inconsistent, while conventional automated approaches may fail to capture the full complexity of defect morphology.

Wafer maps contain two complementary types of visual information: large-scale geometric topology, such as ring- or center-oriented failure patterns, and fine-grained localized die-level texture. A single feature-extraction backbone may not represent both types of information equally well. In addition, many normal or visually uninformative dies can reduce attention to actual defect regions, and real manufacturing conditions may contain superimposed failure patterns that are difficult to classify using conventional single-defect models.

FabMetrics AI addresses this problem through a dual-branch cross-attention hybrid network. A ResNet-50 backbone enhanced with CBAM learns global wafer topology using spatial and channel attention, while EfficientNet-B0 extracts complementary multi-scale fine-grained texture information. Cross-attention dynamically fuses the two feature streams before ten-class classification.

For model development, the project constructs a balanced augmented training corpus of 35,000 wafer-map images, with 3,500 training images per class across Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full, None, and Multi-Defect. To avoid augmentation leakage, source-level training, validation, and test separation is performed before augmentation, and only the training partition is augmented. The project further integrates OpenCV-based defect localization, dashboard visualization, authentication, PDF inspection reporting, and an auxiliary Cleanroom AI Tutor.

OBJECTIVES

- To construct a balanced augmented training corpus of 35,000 wafer-map images, with 3,500 training images for each of 10 classes.
- To perform source-level train/validation/test separation before augmentation so that augmented derivatives do not leak into validation or test data.
- To develop a dual-branch architecture using ResNet-50 + CBAM for global topology and EfficientNet-B0 for fine-grained multi-scale texture extraction.
- To fuse the two branch representations using cross-attention for dynamic feature interaction and weighting.
- To use focal loss for difficult examples and Stochastic Weight Averaging (SWA) as part of the model-training strategy.
- To classify ten wafer-map categories, including a Multi-Defect class for controlled superimposed defect patterns.
- To evaluate the system using Macro F1-score, accuracy, precision, recall, class-wise metrics, confusion matrix, and inference latency.
- To visualize defect regions through OpenCV-based contour and bounding-box overlays.
- To integrate the trained model with an interactive dashboard, authentication, automated PDF inspection reporting, and a supporting Cleanroom AI Tutor.
- To validate the contribution of individual architecture components through controlled baseline and ablation experiments.

INTRODUCTION

Semiconductor fabrication involves multiple tightly controlled processes such as photolithography, etching, deposition, thermal processing, chemical-mechanical planarization, and automated wafer handling. Abnormalities occurring during these stages may produce recognizable spatial patterns in wafer test maps. Automated recognition of such patterns can assist engineers in identifying recurring process problems and improving inspection efficiency.

FabMetrics AI is an end-to-end wafer-map defect recognition and inspection platform designed around a proposed dual-branch cross-attention hybrid network. The first branch uses ResNet-50 with Convolutional Block Attention Module (CBAM) to learn global wafer structure while emphasizing important spatial regions and feature channels. This branch produces a 2048-dimensional feature representation. The second branch uses EfficientNet-B0 to extract multi-scale fine-grained die-texture information and produces a 1280-dimensional representation. The two feature streams are combined through cross-attention so that complementary information can be dynamically weighted before classification.

The project targets ten classes: Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full, None, and Multi-Defect. A balanced augmented training corpus of 35,000 images is constructed with 3,500 training images per class. The Multi-Defect category represents controlled superposition of constituent defect patterns and is treated carefully because synthetic combinations must ultimately be validated against genuine multi-failure wafer maps.

A major methodological concern in augmented datasets is data leakage. Therefore, the project uses a source-level split before augmentation. Original source images are assigned to training, validation, and test groups first; augmentation is applied only to the training group. This prevents transformed versions of the same source image from appearing in both training and evaluation partitions.

The final classifier uses focal loss to emphasize hard-to-classify examples rather than simply to compensate for class-count imbalance, since the augmented training corpus is balanced. SWA is included as part of the training strategy. Beyond classification, OpenCV thresholding, masking, contour detection, and bounding-box overlays provide visual defect localization. FastAPI-based services connect the model to an interactive web dashboard, PDF report generation, authentication, and a Cleanroom AI Tutor. The current project-reported performance of the full model is 97.84% validation Macro F1-score, 98.92% accuracy, and approximately 16.2 ms inference latency; these results must be accompanied by the exact evaluation split and hardware configuration in the final report.

METHODOLOGY

- Collect, clean, and organize source wafer maps into the ten target classes.
- Perform a source-level train/validation/test split before any augmentation to prevent data leakage.
- Apply class-wise augmentation only to the training partition until the balanced training corpus reaches 3,500 images per class (35,000 training images total).
- Resize wafer maps to 224×224 pixels and convert them into normalized model inputs.
- Process each input in parallel through ResNet-50 + CBAM (2048-D global topology features) and EfficientNet-B0 (1280-D multi-scale texture features).
- Fuse both feature representations through a cross-attention module for dynamic feature weighting.
- Use a focal-loss classification head and Stochastic Weight Averaging (SWA) during training, then generate the ten-class prediction and confidence score.
- Apply OpenCV-based thresholding, masking, contour detection, and bounding-box visualization for defect localization.
- Present prediction, confidence, visualization, and PDF inspection-report options through the FastAPI-based dashboard.

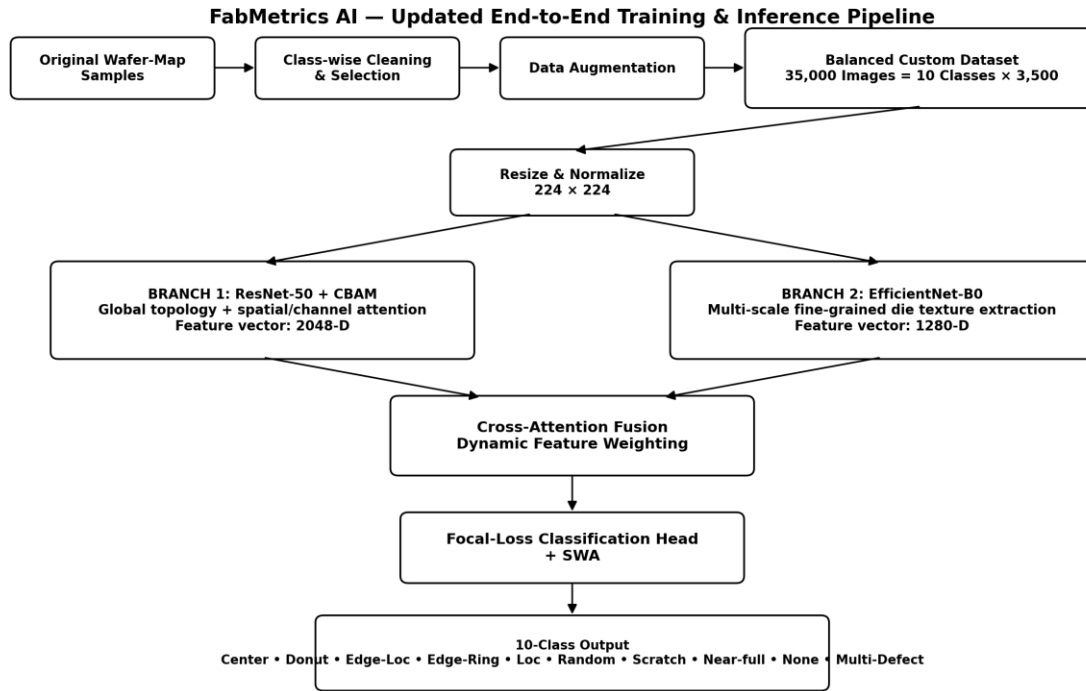


Figure 1. Proposed FabMetrics AI model architecture and processing pipeline

Software/Tools: Python 3.10+, PyTorch 2.x, Torchvision, OpenCV, Pillow, NumPy, Pandas, Scikit-learn, FastAPI, Uvicorn, HTML5, JavaScript, Tailwind CSS, ReportLab, and Google Gemini 2.5 Flash API.

IMPORTANCE OF THE PROJECT IN CONTEXT OF CURRENT SCENARIO

Modern semiconductor manufacturing requires high-throughput and consistent quality control. As device complexity and wafer volumes increase, manual interpretation of wafer maps becomes difficult to scale. Automated defect-pattern recognition can support faster screening and provide a repeatable method for identifying abnormal spatial patterns.

The project is relevant because it combines semiconductor manufacturing with current developments in deep learning, computer vision, attention mechanisms, and Industry 4.0 inspection systems. Instead of relying on a single feature extractor, FabMetrics AI combines global topology and fine-grained texture information through a dual-branch architecture and cross-attention. This is particularly useful for wafer maps because some defect modes are primarily geometric while others depend on localized die-level structures.

The balanced augmented training strategy also addresses the practical challenge of obtaining sufficient examples for all target classes. At the same time, the project explicitly separates source-level evaluation data before augmentation to reduce the risk of inflated performance caused by data leakage. The inclusion of a Multi-Defect category explores a more difficult inspection scenario in which multiple defect patterns may occur simultaneously.

From an application perspective, the project goes beyond standalone classification by integrating prediction, confidence display, OpenCV-based localization, PDF inspection reporting, and an interactive dashboard. Such a workflow demonstrates how an AI model can be converted into a usable inspection-support platform. However, the reported results remain project-level results and require external validation on independent, non-augmented real-fab data before any production-level claim can be made.

TIME SCHEDULE OF ACTIVITIES

Work will be done by Mid Viva:

- Literature review and finalization of problem statement, target classes, and evaluation metrics.
- Collection, cleaning, and organization of source wafer-map data.
- Implementation of source-level train/validation/test separation and class-wise augmentation pipeline.
- Construction of the balanced 35,000-image training corpus.
- Implementation of ResNet-50 + CBAM and EfficientNet-B0 branches.
- Initial training, baseline experiments, and preliminary performance evaluation.

Work will be done by End Viva:

- Implementation and optimization of cross-attention fusion, focal-loss head, and SWA training.
- Completion of ten-class and Multi-Defect experiments with class-wise metrics and confusion matrix.
- Controlled ablation study for ResNet-50, CBAM, EfficientNet-B0, cross-attention, focal loss, and SWA.
- Integration of OpenCV defect localization and bounding-box visualization.
- Completion of FastAPI backend, interactive dashboard, authentication, PDF reporting, and Cleanroom AI Tutor.
- Final testing, latency measurement with documented hardware, external-data validation where available, and preparation of final report/presentation.

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